

* Cost function

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (\overset{\text{Predicted}}{h_0(x)^{(i)}} - \underset{\text{Error}}{y^{(i)}})^2$$

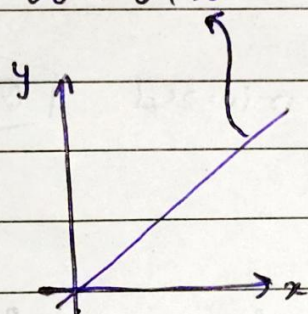
$\Downarrow$
Mean Squared Errors.

Find Aim what we need to solve.

Minimize. $J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_0(x)^{(i)} - y^{(i)})^2 \quad \Downarrow \Downarrow$

1. $h_0(x) = \theta_0 + \theta_1 x$

$\boxed{\theta_0 = 0}$

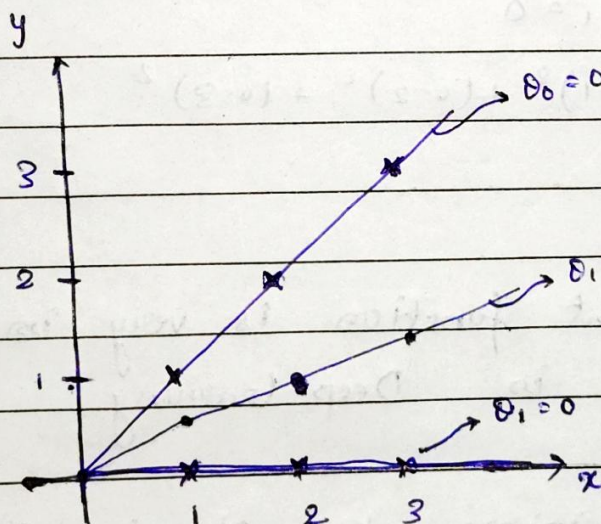


$h_0(x) = \theta_1 x$

DATASET

x	y
1	1
2	2
3	3

Note: $m \rightarrow$ no. of points.



$\theta_0 = 0$

$h_0(x) = \theta_1 x$

let $\theta_1 = 1$ {slope}

$h_0(x) = 1 \quad x = 1$

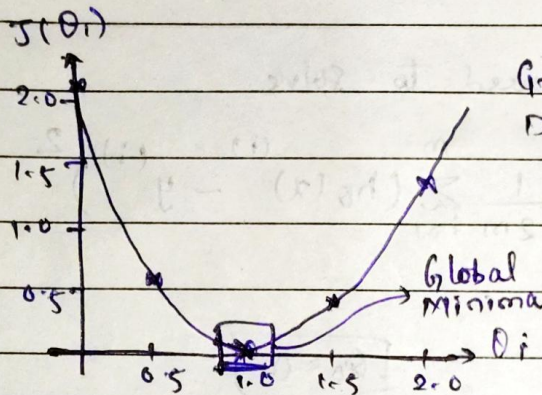
$h_0(x) = 2 \quad x = 2$

$h_0(x) = 3 \quad x = 3$

$$J(\theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

$$= \frac{1}{2 \times 3} [(1-1)^2 + (2-2)^2 + (3-3)^2]$$

$$J(\theta_1) \approx 0 \quad \text{if } \theta_1 = 1$$



$$h_{\theta}(x) = \theta_1 x$$

$$\text{let } \theta_1 = 0.5$$

$$h_{\theta}(x) = 0.5 \quad \text{if } x = 1$$

$$h_{\theta}(x) = 1 \quad \text{if } x = 2$$

$$h_{\theta}(x) = 1.5 \quad \text{if } x = 3$$

Error has been minimized 1.0 or 0.0

$$\text{if } \theta_1 = 0.5$$

$$J(\theta_1) = \frac{1}{2 \times 3} [(0.5-1)^2 + (1-2)^2 + (1.5-3)^2]$$

$$J(\theta_1) \approx 0.58$$

$$\text{if } \theta_1 = 0$$

$$J(\theta_1) = \frac{1}{2 \times 3} [(0-1)^2 + (0-2)^2 + (0-3)^2]$$

$$J(\theta_1) \approx 2.5$$

Note: Gradient Descent function is very important and its also used in Deep learning.

we achieve global minima by changing the θ_1 and θ_0 .
If we get this point that basically indicate we achieved the best fit line.